

Makeup Sales Analysis Project Report

1. Project Objective

The primary objective of this project is to analyze makeup sales data to uncover meaningful business insights such as revenue trends, product performance, customer preferences, and regional sales distribution.

The goal is to transform raw sales data into actionable insights using data analytics techniques and present them through an interactive dashboard to support data-driven decision-making in the cosmetics industry.

2. Tools & Technologies Used

- Google Colab — Data Cleaning, EDA, Feature Engineering
- Pandas — Data manipulation and preprocessing
- Matplotlib & Seaborn — Data visualization
- MySQL — SQL-based analytics
- Power BI — Interactive dashboard development

3. Data Cleaning Process

The dataset was first cleaned in Google Colab to ensure consistency, accuracy, and readiness for analysis.

Key Steps Performed:

- Verified dataset structure and checked missing values
- Confirmed no null values in dataset

	0
Sale_ID	0
Date	0
Brand	0
Product_Type	0
Country	0
Sales_Channel	0
Payment_Method	0
Price_USD	0
Units_Sold	0
Revenue_USD	0
dtype: int64	

- Converted `Date` column to proper datetime format

- Removed duplicate records
- Standardized categorical variables (Brand, Product_Type, Country, Sales_Channel, Payment_Method)
- Ensured consistent text formatting for better grouping and analysis
- Exported cleaned dataset as `clean_makeup_sales.csv`

Outcome:

A clean, structured dataset ready for advanced analysis and modeling.

4. Exploratory Data Analysis (EDA)

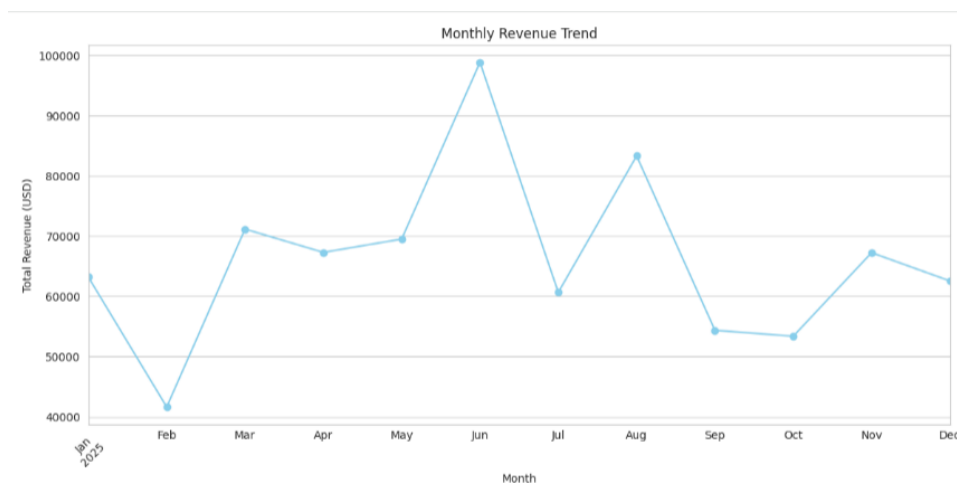
EDA was performed to understand patterns, trends, and relationships within the dataset.

Key Analyses Conducted:

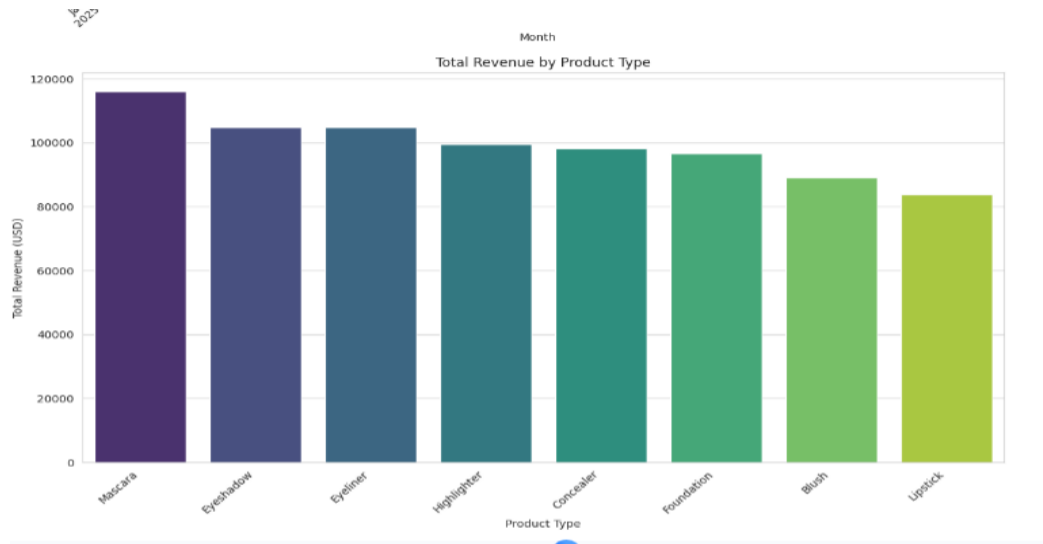
- Monthly revenue trend analysis
- Product category performance evaluation
- Brand-wise revenue comparison
- Country-wise sales distribution
- Sales channel performance analysis

Key Visualizations:

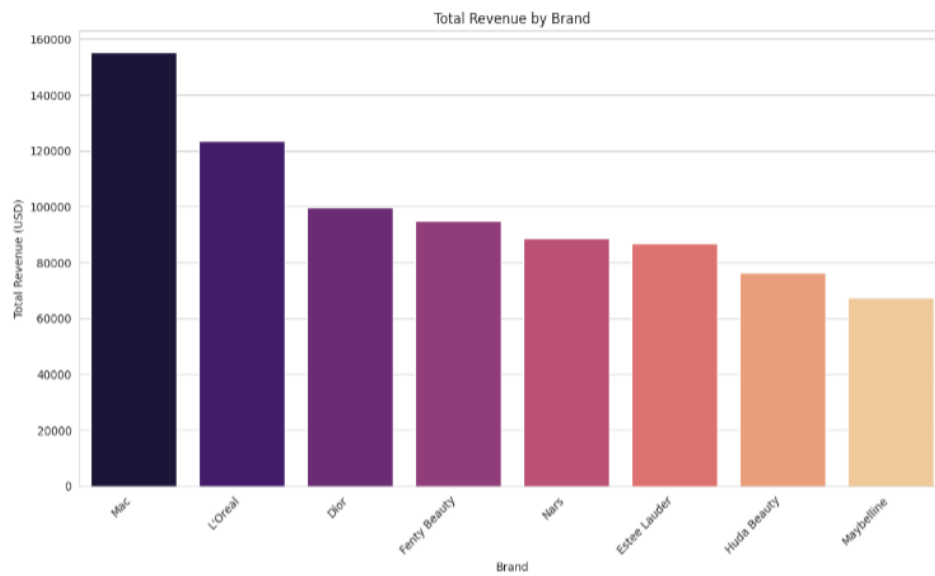
- Monthly Revenue Trend (Line Chart)



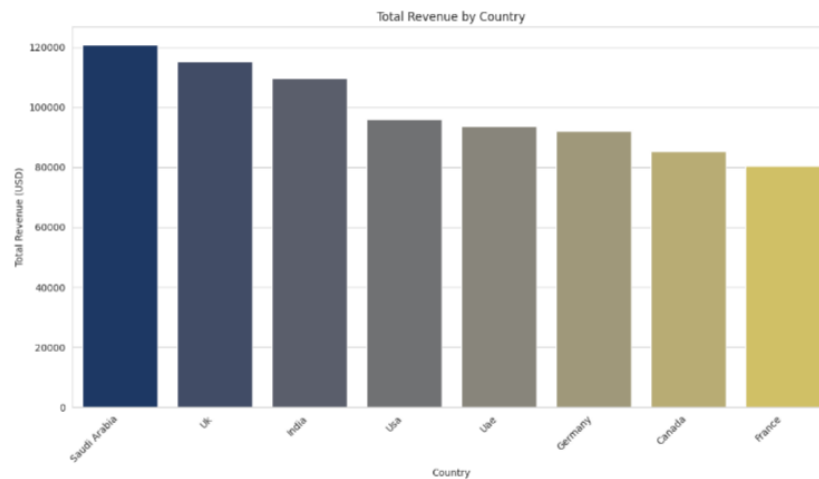
- Product Type Performance (Bar Chart)



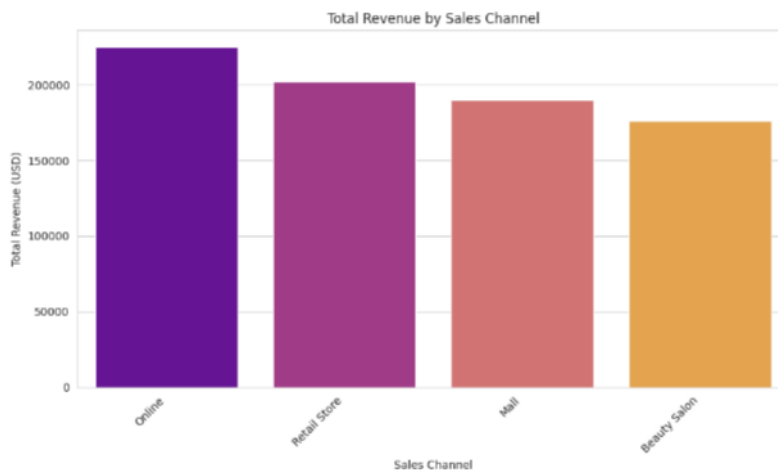
- Brand Performance (Bar Chart)



- Country-wise Revenue Distribution



- Sales Channel Comparison



5. Feature Engineering

To enhance analytical depth, new features were created from existing data.

Features Created:

- Recalculated Revenue = Price_USD × Units_Sold (validation check)
- Year extraction from Date
- Month Name and Month Number
- Day of Week analysis

Additional Notes:

Customer Lifetime Value (CLV) analysis was not performed due to the absence of a unique Customer_ID field.

6. SQL-Based Analysis

The cleaned dataset was imported into MySQL for structured querying and advanced business analysis.

Key SQL Analyses Performed:

- Top 10 Products by Revenue

Product_Type	Brand	Total_Revenue
Eyeliner	Mac	29470.05
Blush	Mac	25403.80
Highlighter	Mac	22258.94
Mascara	Huda Beauty	21740.03
Concealer	Mac	20419.64
Foundation	Nars	19879.07
Lipstick	Mac	19677.32
Highlighter	L'Oreal	19133.65
Mascara	Dior	18550.02

- Monthly Sales Trend

Month	Monthly_Revenue
2025-01	63184.05
2025-02	41654.67
2025-03	71169.11
2025-04	67298.71
2025-05	69506.68
2025-06	98829.22
2025-07	60675.03
2025-08	83296.77
2025-09	54373.64

- Country-wise Performance

Country	Total_Revenue	Total_Units
Saudi Arabia	120778.08	1697
Uk	115359.56	1760
India	109638.23	1740
Usa	96005.51	1458
Uae	93567.94	1491
Germany	92125.73	1653
Canada	85349.39	1602
France	80364.06	1307

- Best Selling Product Types

Product_Type	Total_Units_Sold
Mascara	1900
Eyeshadow	1716
Eyeliners	1646
Lipstick	1592
Highlighter	1526
Foundation	1523
Blush	1453
Concealer	1352

- Payment Method Analysis

Payment_Method	Total_Transactions	Total_Revenue
Digital Wallet	180	290527.71
Card	186	282921.96
Cash	134	219738.83

- Sales Channel Performance

Sales_Channel	Total_Revenue	Total_Sales
Online	224882.60	139
Retail Store	202425.43	117
Mall	189931.46	123
Beauty Salon	175949.01	121

- Top Brands by Revenue

Brand	Total_Revenue
Mac	155245.81
L'Oreal	123593.20
Dior	99810.99
Fenty Beauty	95001.67
Nars	88792.62
Estee Lauder	86983.48
Huda Beauty	76357.17
Mavbelline	67403.56

- Daily Sales Trend

Date	Daily_Revenue
2025-01-01	7555.75
2025-01-02	2339.10
2025-01-05	2322.97
2025-01-09	6548.05
2025-01-10	633.65
2025-01-13	1295.03
2025-01-14	4137.34
2025-01-15	6064.80
2025-01-16	5579.89

- High Value Transactions (Premium Sales)

Sale_ID	Brand	Product_Type	Revenue_USD
390	Huda Beauty	Eyeline	5987.50
78	L'Oreal	Mascara	5328.18
237	Dior	Eyeline	5281.86
388	Fenty Beauty	Concealer	5229.50
51	Fenty Beauty	Concealer	5220.00
41	Mac	Blush	5151.36
241	Dior	Highlighter	5004.56
494	Mac	Eyeshadow	4897.40
11	Huda Beauty	Concealer	4870.80
356	L'Oreal	Highlighter	4819.00

- Average Order Value

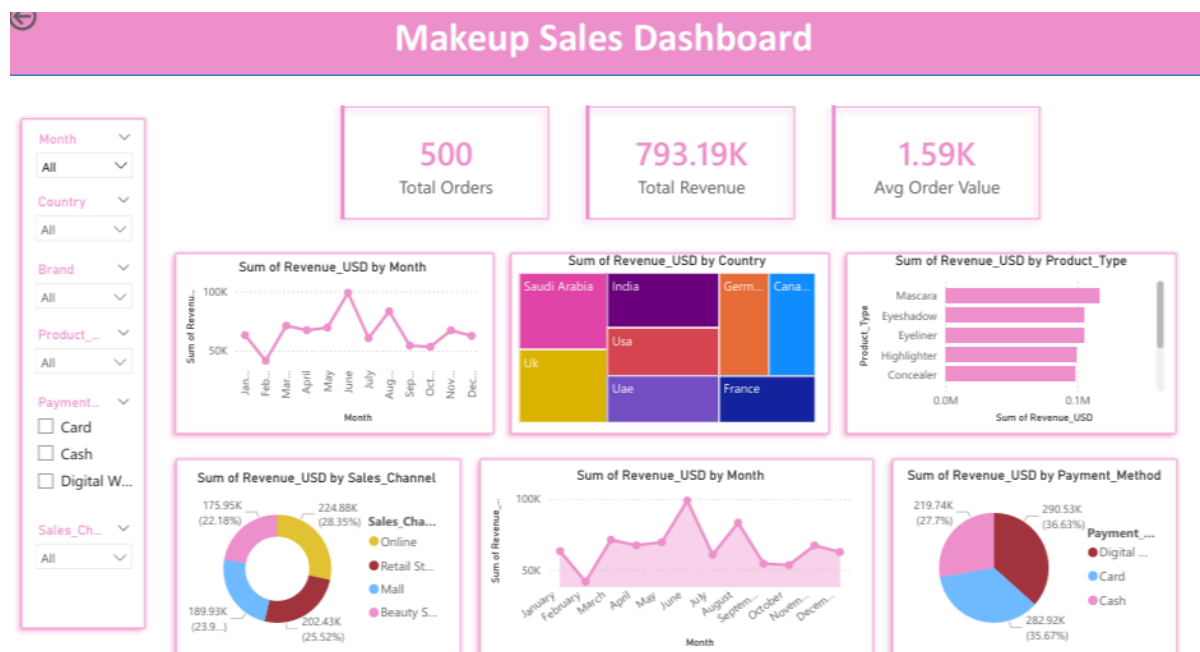
Avg_Order_Value
1586.377000

Outcome:

Enabled structured, scalable, and business-oriented insights from raw transactional data.

7. Dashboard Development

An interactive dashboard was created using Power BI to visually present key business metrics.



8. Key Insights

- Lipstick and foundation categories contributed the highest revenue share
- Online sales channel consistently outperformed offline channels
- Certain countries contributed significantly higher revenue than others
- Strong seasonal trends were observed in monthly sales performance
- A small number of top brands dominated overall revenue contribution

9. Business Recommendations

Based on the analysis, the following strategic recommendations are proposed:

1. Focus on High-Performing Products

Increase inventory and marketing efforts for top-selling categories such as lipstick and foundation, as they generate the highest revenue.

2. Strengthen Regional Strategy

Allocate more marketing budget and distribution resources to high-performing countries while developing targeted campaigns for underperforming regions.

3. Expand Online Sales Channels

Since online sales outperform offline channels, strengthen e-commerce platforms, improve digital marketing, and enhance online customer experience.

4. Optimize Payment Options

Promote preferred payment methods to improve transaction efficiency and reduce checkout friction.

5. Seasonal Marketing Campaigns

Leverage monthly sales trends to launch seasonal promotions during peak sales periods to maximize revenue.

6. Brand Partnership Strategy

Focus on strengthening partnerships with top-performing brands while exploring opportunities to onboard emerging brands with growth potential.

10. Conclusion

This project demonstrates a complete end-to-end data analytics workflow, including:

- Data Cleaning and Preparation
- Exploratory Data Analysis (EDA)

- Feature Engineering
- SQL-Based Business Analysis
- Interactive Dashboard Development

The project successfully transforms raw transactional data into meaningful business insights that can support strategic decision-making in the cosmetics industry.